Redox

Introduction

1) Hydrogen economy : Use of liq Hydrogen as fuel.
2) Ozone hole : depletion of ozone layer due to CFCs.

These are redox reaction.

Classical idea of Redox

Oxidation

① addition of oxygen
② addition of more En atom
③ Removal of Hydrogen
④ Removal of electropositive element

① $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$

$S(s) + O_2(g) \rightarrow SO_2(g)$

③ $CH_4(g) + O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

addition of oxygen → oxidation
but hydrogen is also removed.

$2H_2S(g) + O_2(g) \rightarrow 2S(s) + 2H_2O(l)$

② $Mg(s) + F_2(g) \rightarrow MgF_2(s)$

$Mg(s) + Cl_2(g) \rightarrow MgCl_2(s)$

$Mg(s) + S(s) \rightarrow MgS(s)$

④ $2K_4[Fe(CN)_6](aq) + H_2O_2(aq) \rightarrow 2K_3[Fe(CN)_6](aq) + 2KOH(aq)$

Removal of more electropositive element

Reduction

① removal of oxygen
② removal of more En atom
③ addition of Hydrogen
④ addition of electropositive element

$2HgO(s) \xrightarrow{\Delta} 2Hg(l) + O_2(g)$

$2FeCl_3(aq) + H_2(g) \rightarrow 2FeCl_2(aq) + 2HCl(aq)$

$CH_2=CH_2(g) + H_2(g) \rightarrow CH_3-CH_3(g)$

$2HgCl_2(aq) + SnCl_2(aq) \rightarrow Hg_2Cl_2(s) + SnCl_4(aq)$

↓

Here oxidation as well as reduction both occurs.

↓

These type of rxn is called redox rxn.

Que In the reaction given below identify species undergoing oxidation and reduction:-

① $H_2S(g) + Cl_2(g) \rightarrow 2HCl(g) + S(s)$

② $3Fe_3O_4(s) + 8Al(s) \rightarrow 9Fe(s) + 4Al_2O_3(s)$

③ ~~$2Na(s) + H_2(g) \rightarrow 2NaH(s)$~~

H_2S → oxidize (bcoz more En atom Cl is added to H)
(or more electropositive element H is removed from S)

Al → oxidize (oxygen added) Fe_3O_4 → Reduced (oxygen removed)

Electronic Concept or Modern concept of Redox

Trick → OIL RIG

→ oxidation is loss of electron

→ Reduction is gain of e^-

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s)$ (loss of $2e^-$; gain of $2e^-$; $Na^0 \to Na^{+1}$, $Cl^0 \to Cl^{-1}$)

$2Na(s) + O_2(g) \longrightarrow Na_2O(s)$ (loss of $2e^-$; gain of $2e^-$; $Na^0 \to Na^{+1}$, $O^0 \to O^{-2}$)

$2Na(s) + S(s) \longrightarrow Na_2S(s)$ (loss of $2e^-$; gain of $2e^-$; $Na^0 \to Na^{+1}$, $S^0 \to S^{-2}$)

Half Reaction

write loss and gain of electron separately.

$2Na(s) \longrightarrow 2Na^+(g) + 2e^-$] loss of e^- → oxidation; Half reaction; oxidation half reaction

$2e^- + Cl_2(g) \longrightarrow 2Cl^-(g)$] Half rxn; gain of e^- → Reduction; Reduction half reaction

Oxidizing agent (oxidant): oxidizes others & reduces themselves

RIG (acceptor of e^-) (decrease in charge)

$Cu(s) + 4H^+ + NO_3^- \to Cu^{+2} + H_2O + NO_2$

NO_3^-: reduces itself & oxidizes copper → oxidizing agent

Reducing agent (Reductant): Reduces other & oxidizes themselves.

OIL (Donor of e^-) (increase in charge)

$Sn^{+2} + Fe^{+3} \longrightarrow Sn^{+4} + Fe^{+2}$

Sn^{+2}: oxidizes itself; Reduces Fe^{+3}; Reducing agent

Que Identify where the following is a redox reaction:-

$2Na^0(s) + H_2^0(g) \longrightarrow 2Na^{+1}H^{-1}(s)$ } Redox

Ox; Red

Que Identify oxidizing & Reducing agent in the following:-

$Fe^{+2}O + Mn^{+7}O_4^- + H^+ \longrightarrow Fe_2^{+3}O_3 + Mn^{+2}Cl_2 + H_2O$.

(Reduction; Oxdn)

FeO → Oxdn → Reducing Agent.

MnO_4^- → Redn → oxidizing agent.

Competitive electron transfer reaction (2)

Reactivity series

Please → Potasium
stop → Sodium
Calling → Calcium
me → magnetium
a → aluminium
cute → Carbon
zebra → Zinc
I → Iron
never → nickel
today → Tin
like → Lead
her → Hydrogen
Call → Copper
Smart → silver
goat → Gold
Poor → Platinum

- Higher reactive will replace lower reactive.

(1) Reaction blw Zn and copper nitrate solution.

- $Zn^{0}_{(s)} + Cu(NO_3)_2 \rightarrow Zn(NO_3)_2 + Cu^{0}$
  (Zn: oxidn; Cu: Redn)
  $Cu^{+2}(NO_3)_{2(aq)}$ — Blue color
  $Zn(NO_3)_{2(aq)}$ — Colorless
  $Cu^{0}_{(s)}$ — Colorless

- $Cu_{(s)} + Zn(SO_4)_{(aq)} \rightarrow X$ (no reaction)

- $Cu_{(s)} + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$
  (Cu: oxidn (loss of $2e^-$))
  $Cu_{(s)}$ — Colorless
  $Cu(NO_3)_2$ — Blue color
  Ag: Redn (gain of $2e^-$)

Zinc releases e^- to Cu and Cu releases e^- to Ag. therefore e^- releasing tendency $Zn > Cu > Ag$.

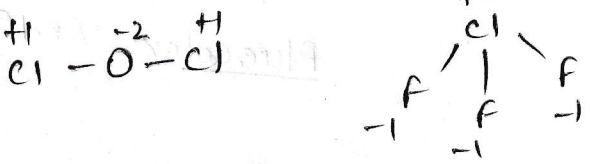
Oxidation Num

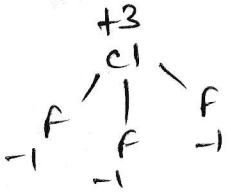
- charge present on an atom in a given molecule.

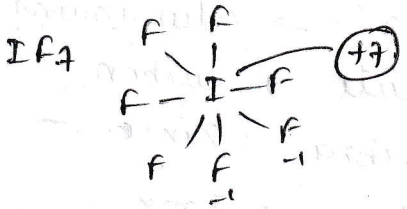
Rule for writing O. Number

- ON of element in elemental state/free state is zero.
  H_2, O_2, N_2, O_3, Cl_2, Fe, Cu, Ni, P_4, S_8, Na, Mg, Al
- alkali metal → (+1), alkaline earth metal → (+2), Al → (+3)
  Li, Na, K, Rb, Cs, Fr
  Be, Mg, Ca, Sr, Ba, Ra
- For ions oxidation number = charge. $Na^{+} = 1$, $Mg^{+2} = 2$, $Al^{+3} = 3$, $Fe^{+3} = 3$, $Cl^{-} = -1$, $O^{2-} = -2$.
- Oxygen — generally −2 oxidation number
  Peroxide (−1) H_2O_2, Na_2O_2
  Superoxide (−1/2) KO_2, RbO_2, CsO_2
  oxyfluoride → (+1), (+2) OF_2, O_2F_2

- Hydrogen (generally +1)
  but in metal hydride (-1)
  eg: LiH, NaH, CaH_2, MgH_2
- Halogen (F, Cl, Br, I) → generally -1
  ↓
  always -1.
  But Cl, Br, I → +ve oxd^n state (when combined with more En atom)







$\overset{+7}{HClO_4}$

$\overset{+5}{HClO_3}$

$\overset{+3}{HClO_2}$

$\overset{+1}{HClO}$

- Radical

		Oxd^n num
X^- (F^-, Cl^-, Br^-, I^-)	halide	-1
O^{-2}	oxide	-2
N^{-3}	nitride	-3
P^{-3}	Phosphide	-3
S^{-2}	Sulphide	-2
OH^-	hydroxide	-1
CN^-	Cyanide	-1
NC^-	isocyanide	-1
SO_3^{-2}	Sulphite	-2
SO_4^{-2}	Sulphate	-2
PO_3^{-3}	Phosphite	-3
PO_4^{-3}	phosphate	-3
NO_2^-	nitrite	-1
NO_3^-	nitrate	-1
$C_2O_4^{-2}$	oxalate	-2
$S_2O_3^{-2}$	thiosulphate	-2
$S_4O_6^{-2}$	tetrathionate.	-2

ClO^-	– Hypochlorite	-1
ClO_2^-	→ Chlorite	-1
ClO_3^-	– Chlorate	-1
ClO_4^-	– Perchlorate	-1
MnO_4^{-2}	→ mangate	-2
MnO_4^-	– Permanganate	-1
CrO_4^{-2}	– chromate	-2
$Cr_2O_7^{-2}$	– dichromate	-2
SCN^-	– thiocyanate	-1
NCS^-	– isothiocynate	-1
NH_4^+	– Ammonium	+1
No^+	– nitrosyl	+1

- Sum of oxidation num of all elements present in a molecule is equal to charge on molecule.

SO_4^{-2} NH_4^+ H_2O.

$(ON)_S + 4(ON)_O = -2$ | $(ON)_N + 4(ON)_H = +1$ | $2\times(ON)_H + (ON)_O = 0$

$[Ag(S_2O_3)_2]^{-3}$

MnO_4^-

~~SO_4^{-2}~~

$Cr_2O_7^{-2}$

NH_4^+

ClO_4^-

$S_2O_3^{-2}$

~~CrO_4^{-2}~~

~~Fe_2O_3~~

$[Fe(CN)_6]^{-3}$

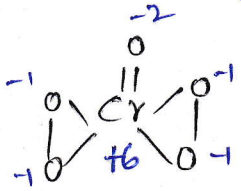
$Ni(CO)_5$

FeO Fe_2O_3 → av. $= \frac{2\times1 + 2\times3}{3} = \frac{8}{3}$

Fe_3O_4 ($\lambda = \frac{8}{3}$) → Fractional oxidation state shows av. oxidation num.

Exceptional oxidation num cases

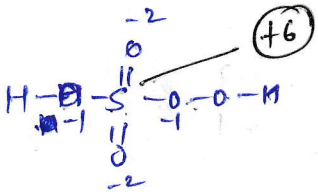
CrO_5



Deep blue butterfly.

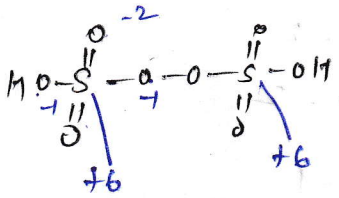
H_2SO_5 → $H_2SO_4 + O$

Caro's acid



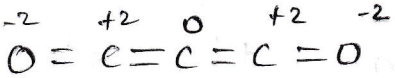
$H_2S_2O_8$

marshall acid



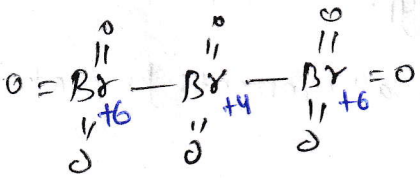
C_3O_2

Carbon suboxide



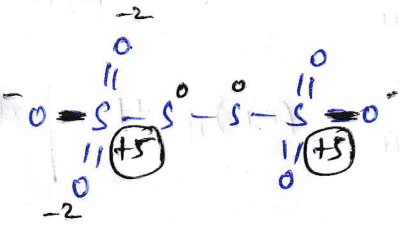
Br_3O_8

tribromooctaoxide



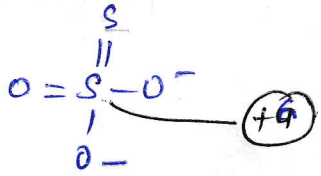
$S_4O_6^{-2}$

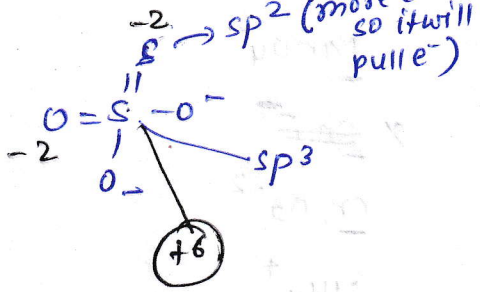
tetrathionate ion



$S_2O_3^{-2}$

thiosulphate ion





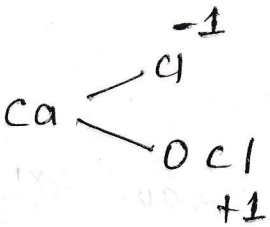
$$\text{av oxd}^n\text{ no} = \frac{6-2}{2}$$

$$= \frac{4}{2}$$

$$= 2$$

Bleaching Powder

$CaOCl_2$



$$\text{av oxd}^n\text{ no} = \frac{1-1}{2} = \underline{0}$$

Brown ring complex

$[Fe(H_2O)_5 \overset{+1}{NO}]\overset{+1}{SO_4}$

Types of Redox Reaction

Combination Rxn : two atoms/molecules combine togather to form a product
- In this rxn atleast one of the reactant must be in elemental form.

$$C_{(s)} + O_{2(g)} \xrightarrow{\Delta} CO_{2(g)}$$

$$3\,Mg_{(s)} + N_{2(g)} \xrightarrow{\Delta} Mg_3N_{2(s)}$$

$$CH_{4(g)} + 2O_{2(g)} \xrightarrow{\Delta} CO_{2(g)} + 2H_2O_{(\ell)}$$

$$H_{2(g)} + O_{2(g)} \xrightarrow{\Delta} 2H_2O_{(\ell)}$$

Decomposition rxn
- Reverse of combination reaction.
- molecule/compd breaks down to form two or more component
- one of the product must be in elemental form.

$$2H_2O_{(\ell)} \xrightarrow{\Delta} 2H_{2(g)} + O_{2(g)}$$

$$2NaH_{(s)} \xrightarrow{\Delta} 2Na_{(s)} + H_{2(g)}$$

$$2KClO_{3(s)} \xrightarrow{\Delta} 2KCl_{(s)} + 3O_{2(g)}$$

$$4LiNO_{3(s)} \xrightarrow{\Delta} 2Li_2O_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

$$\overset{+2\ +4\ -2}{CaCO_{3(s)}} \xrightarrow{\Delta} \overset{+2\ -2}{CaO_{(s)}} + \overset{+4\ -2}{CO_{2(g)}}$$

→ Decomposition rxn — non-Redox
⇓
oxidn state not changing.

Displacement reaction : an atom/ion is replaced by another metal/ion.

$$AB + C \longrightarrow AC + B$$

- metal displacement rxn → metal atom/ion is displaced by another metal.
- non-metal displacement rxn → non metal atom/ion is displaced by another metal or non-metal.

$$CuSO_{4(aq)} + Zn_{(s)} \longrightarrow Cu_{(s)} + ZnSO_{4(aq)}$$

$$V_2O_{5(s)} + 5Ca_{(s)} \xrightarrow{\Delta} 2V_{(s)} + 5CaO_{(s)}$$

$$TiCl_{4(\ell)} + 2Mg_{(s)} \xrightarrow{\Delta} Ti_{(s)} + 2MgCl_{2(s)}$$

$$Cr_2O_{3(s)} + 2Al_{(s)} \xrightarrow{\Delta} Al_2O_{3(s)} + 2Cr_{(s)}$$

$$2Na_{(s)} + 2H_2O_{(\ell)} \longrightarrow 2NaOH_{(aq)} + H_{2(g)}$$

$$Ca_{(s)} + 2H_2O_{(\ell)} \longrightarrow Ca(OH)_{2(aq)} + H_{2(g)}$$

$$Mg_{(s)} + 2H_2O_{(\ell)} \longrightarrow Mg(OH)_{2(s)} + H_{2(g)}$$

$$2Fe_{(s)} + 3H_2O_{(\ell)} \longrightarrow Fe_2O_{3(s)} + 3H_{2(g)}$$

$Zn_{(s)} + 2HCl_{(aq)} \longrightarrow ZnCl_{2(aq)} + H_{2(g)}$

$Mg_{(s)} + 2HCl_{(aq)} \longrightarrow MgCl_{2(aq)} + H_{2(g)}$

$Fe_{(s)} + 2HCl_{(aq)} \longrightarrow FeCl_{2(aq)} + H_{2(g)}$

$2H_2O_{(l)} + 2F_{2(g)} \longrightarrow 4HF_{(aq)} + O_{2(g)}$

$Cl_{2(g)} + 2KBr_{(aq)} \longrightarrow 2KCl_{(aq)} + Br_{2(l)}$

$Cl_{2(g)} + 2KI_{(aq)} \longrightarrow 2KCl_{(aq)} + I_{2(s)}$

$Br_{2(l)} + 2KI_{(aq)} \longrightarrow 2KBr_{(aq)} + I_{2(s)}$

Disproportionation reaction

Rxn in which oxidation & reduction of same element occurs from same oxidation state.

$2\overset{-1}{H_2O_2}_{(aq)} \longrightarrow 2\overset{-2}{H_2O}_{(l)} + \overset{0}{O_2}_{(g)}$

$\overset{0}{P_4}_{(s)} + 3NaOH_{(aq)} + 3H_2O \longrightarrow \overset{-3}{PH_3}_{(g)} + 3\overset{+1}{NaH_2}PO_{2(aq)}$

$\overset{0}{S_8}_{(s)} + 12NaOH_{(aq)} \longrightarrow 4\overset{-2}{Na_2S}_{(aq)} + 2\overset{+2}{Na_2S_2O_3}_{(aq)} + 6H_2O$

$\overset{0}{Cl_2}_{(g)} + 2NaOH_{(aq)} \xrightarrow{\text{cold \& dilute}} \overset{+1}{NaOCl}_{(aq)} + \overset{-1}{NaCl}_{(aq)} + H_2O_{(l)}$

$Cl_2^0 + 2NaOH \xrightarrow{\text{(hot \& conc)}} NaCl^{-1} + NaClO_3^{+5} + H_2O$

Que Identify whether it is a disproportionation reaction.

$2\overset{+7}{KClO_3} \xrightarrow{\text{oxidation}} 2KCl^{-1} + 3O_2^0$ (Red)

✗ disprop ✗ → same element + oxdn + Redn

$\overset{-3}{N}H_4\overset{+3}{N}O_2 \longrightarrow N_2^0 + 2H_2O$ ✗ disprop ✗ → same oxdn no → oxdn, Redn

$4\overset{+5}{K}ClO_3 \longrightarrow 3\overset{+7}{K}ClO_4 + K\overset{-1}{Cl}$ ✓ disprop

$F_2^0 + NaOH(dil) \rightarrow NaF^{-1} + OF_2^{+1}$ ✗

$F_2 + NaCl(conc) \rightarrow NaF + O_2$ ✗

$\overset{+6}{MnO_4^{-2}} \longrightarrow \overset{+7}{MnO_4^-} + \overset{+4}{MnO_2}$ ✓

O

R

$\overset{+1}{H_3PO_2} \longrightarrow \overset{-3}{PH_3} + \overset{+3}{H_3PO_3}$ ✓

$H_3PO_3 \longrightarrow PH_3 + H_3PO_4$ ✓

$HClO \longrightarrow HCl + HClO_2$

$HClO_2 \longrightarrow HCl + HClO_3$

$HClO_3 \longrightarrow HCl + HClO_4$

$HNO_2 \longrightarrow NO + NNO_3$

Comproportionation Reaction

- Reverse of disprop.
- reaction in which element changes from two different oxdn state (one lower & other higher) to same O.Num.

$\overset{-3}{NH_4}\overset{+3}{NO_2} \longrightarrow \overset{0}{N_2} + 2H_2O$

Que Why the following rxn show different product

$Pb_3O_4 + 8HCl \longrightarrow 3PbCl_2 + Cl_2 + 4H_2O$

$Pb_3O_4 + 4NNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2 + 2H_2O$

Pb_3O_4 : $\overset{+4}{PbO_2}$: PbO

↓ Unstable

↓ Inert pair effect

So $Pb^{+4} \longrightarrow Pb^{+2}$ (Self Reduction)

So, it will oxidize others.

$Cl^- \longrightarrow Cl_2$

Pb_3O_4 : PbO : PbO_2

$PbO \longrightarrow Pb(NO_3)_2$

PbO_2 → O.A

HNO_2 O.A

↓ So no rxn

Remaing $Pb^{+2} \longrightarrow PbCl_2$

Balancing of Redox Reaction

(1) atom balance &
(2) charge balance

→ ion electron method
पहले atom balance
then charge balance

→ oxidation num method
पहले charge balance
then
atom balance.

(Basic medium)

$$Zn + NO_3^- \longrightarrow Zn^{+2} + NH_4^+$$

Step1: seperate into oxidation half & reduction half.

OH $\quad Zn \longrightarrow Zn^{+2}$

RH $\quad \overset{+5}{N}O_3^- \longrightarrow \overset{-3}{N}H_4^+$

Step2 Balance atoms other than O, H.

already balanced

Step3 Balance O by H_2O, & H by H^+

$$Zn \longrightarrow Zn^{+2}$$

$$10H^+ + NO_3^- \longrightarrow \underset{4H}{NH_4^+} + \underset{6H}{3H_2O}$$

Step4 in basic medium Remove H^+ by adding OH^-

$Zn \longrightarrow Zn^{+2}$

$$10H^+ + NO_3^- \longrightarrow NH_4^+ + 3H_2O + 10OH^-$$

$10OH^-$

$$10H_2O + NO_3^- \longrightarrow NH_4^+ + 3H_2O + 10OH^-$$

$$7H_2O + NO_3^- \longrightarrow NH_4^+ + 10OH^-$$

chargebalane.

OA	OP
RA	RR

Step5 ~~Balance charge~~

$$Zn \longrightarrow Zn^{+2} + 2e^- \quad \text{Oxdn}$$

$$8e^- + 7H_2O + \overset{+5}{N}O_3^- \longrightarrow \overset{-3}{N}H_4^+ + 10OH^- \quad \text{Redn}$$

Step6 add OH & RH also Cancel e^-.

$$(Zn \longrightarrow Zn^{+2} + 2e^-) \times 4$$

$$8e^- + 7H_2O + NO_3^- \longrightarrow NH_4^+ + 10OH^-$$

$$4Zn \longrightarrow 4Zn^{+2} + 8e^-$$

$$8e^- + 7H_2O + NO_3^- \longrightarrow NH_4^+ + 10OH^-$$

$$4Zn + 7H_2O + NO_3^- \longrightarrow 4Zn^{+2} + NH_4^+ + 10OH^-$$

$Fe^{+2} + \overset{+6}{Cr_2O_7^{-2}} \longrightarrow Fe^{+3} + Cr^{+3}$ (Balance in acidic medium)

Step 1. Seperate OH & RH

$Fe^{+2} \longrightarrow Fe^{+3}$

$Cr_2O_7^{-2} \longrightarrow Cr^{+3}$

Step 2 Balance atoms other than O & H

$Fe^{+2} \longrightarrow Fe^{+3}$

$Cr_2O_7^{-2} \longrightarrow 2Cr^{+3}$

Step 3 Balance O by H_2O & H by H^+

$Fe^{+2} \longrightarrow Fe^{+3}$

$14H^+ + Cr_2O_7^{-2} \longrightarrow 2Cr^{+3} + 7H_2O$

Step 4 → atom balanced

acidic medium

no need ~~for~~ to ~~add~~ remove H^+ by adding OH^-

$Fe^{+2} \longrightarrow Fe^{+3} + e^-$ oxidn

$6e^- + 14H^+ + \overset{+6}{Cr_2O_7^{-2}} \longrightarrow 2Cr^{+3} + 7H_2O$ Redn

Step 5 charge Balance →

$6Fe^{+2} \longrightarrow 6Fe^{+3} + 6e^-$

$6e^- + 14H^+ + Cr_2O_7^{-2} \longrightarrow 2Cr^{+3} + 7H_2O$

Step 6 → add OH & RH and cancel e^-

$6Fe^{+2} + 14H^+ + Cr_2O_7^{-2} \longrightarrow 6Fe^{+3} + 2Cr^{+3} + 7H_2O$

Que $\overset{+7}{KMnO_4} + \overset{-1}{KI} + HCl \xrightarrow{\text{acidic medium}} \overset{+2}{MnCl_2} + \overset{0}{I_2} + KCl + H_2O$

Step1 → Remove spectator ion & convert into ionic form.

$$MnO_4^- + I^- \longrightarrow MnCl_2 + I_2$$

Step2 OH & RH

$\overset{+7}{MnO_4^-} \longrightarrow \overset{+2}{MnCl_2}$ RH

$I^- \longrightarrow I_2$ OH

Step 3 Balance atom other than O & H

$MnO_4^- \longrightarrow MnCl_2$

$2I^- \longrightarrow I_2$

step 3| O balance by H_2O H by H^+

$8H^+ + MnO_4^- \longrightarrow MnCl_2 + 4H_2O$

$2I^- \longrightarrow I_2$

step 5 balance charge

$5e^- + 8H^+ + \overset{+7}{MnO_4^-} \longrightarrow \overset{+2}{MnCl_2} + 4H_2O$ Redn

$2I^- \longrightarrow I_2^0 + 2e^-$ oxdn

step 6

Redn → × 2

oxdn → × 5

$10e^- + 16H^+ + 2MnO_4^- \rightarrow 2MnCl_2 + 8H_2O$

$10I^- \longrightarrow 5I_2 + 10e^-$

$2MnO_4^- + 10I^- + 16H^+ \rightarrow 2MnCl_2 + 5I_2 + 8H_2O$

step 7: Place back spectators

$2KMnO_4 + 10KI + 16HCl \longrightarrow 2MnCl_2 + 5I_2 + 8H_2O$

$12KCl$

Qu1

$$\overset{+7}{KMnO_4} + \overset{+3}{H_2C_2O_4} + H_2SO_4 \text{ (acidic medium)} \longrightarrow K_2SO_4 + \overset{+2}{MnSO_4} + \overset{+4}{CO_2} + H_2O$$

(7)

Step 1 Remove spectator ion

$$MnO_4^- + C_2O_4^{-2} \longrightarrow Mn^{+2} + CO_2$$

Step 2 OH & RH

$$\overset{+7}{MnO_4^-} \longrightarrow \overset{+2}{Mn} \quad \text{RH}$$

$$\overset{+3}{C_2O_4^{-2}} \longrightarrow \overset{+2}{CO_2} \quad \text{OH}$$

Step 3 Balance other than O & H

$$MnO_4^- \longrightarrow \overset{+2}{Mn}$$

$$C_2O_4^{-2} \longrightarrow 2CO_2$$

Step 4 O → H_2O, H → H^+

$$8H^+ + MnO_4^- \longrightarrow \overset{+2}{Mn} + 4H_2O$$

$$C_2O_4^{-2} \longrightarrow 2CO_2$$

Step 5 Charge balance.

$$5e^- + 8H^+ + \overset{+7}{MnO_4^-} \longrightarrow \overset{+2}{Mn^{+2}} + 4H_2O \quad \text{Red}^n$$

$$\overset{+3}{C_2O_4^{-2}} \longrightarrow \overset{+4}{2CO_2} + 2e^- \quad \text{oxd}^n$$

Step 6 add & cancel e^-

$\text{Red}^n \times 2$

$\text{oxd}^n \times 5$

$$10e^- + 16H^+ + 2MnO_4^- \longrightarrow 2Mn^{+2} + 8H_2O$$

$$5C_2O_4^{-2} \longrightarrow 10CO_2 + 10e^-$$

$$2MnO_4^- + 5C_2O_4^{-2} + 16H^+ \longrightarrow 2Mn^{+2} + 10CO_2 + 8H_2O$$

(10H)

$$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \longrightarrow 2MnSO_4 + 10CO_2 + 8H_2O + K_2SO_4$$

Oxidation Num method

$MnO_4^- + C_2O_4^{-2} + H^+ \longrightarrow CO_2 + Mn^{+2} + H_2O$

S-1 write oxdn no of element & cross multiply with no of e^- loss/gain. in Reactants only.

$5e^-$

$\overset{+7}{MnO_4^-} + \overset{+3}{C_2O_4^{-2}} + H^+ \longrightarrow \overset{+4}{CO_2} + Mn^{+2} + H_2O$

$2e^-$

$2MnO_4^- + 5C_2O_4^{-2} + H^+ \longrightarrow CO_2 + Mn^{+2} + H_2O$

step2 Balance atoms other than O, H

$2MnO_4^- + 5C_2O_4^{-2} + H^+ \longrightarrow 10CO_2 + 2Mn^{+2} + H_2O$

$2MnO_4^- + 5C_2O_4^{-2} + H^+ \longrightarrow 10CO_2 + 2Mn^{+2} + 8H_2O$

step3 $O \rightarrow H_2O$, $H \rightarrow H^+$

$2MnO_4^- + 5C_2O_4^{-2} + 16H^+ \longrightarrow 10CO_2 + 2Mn^{+2} + 8H_2O.$

---

acidic med $Cr_2O_7^{-2} + SO_3^{-2} \longrightarrow Cr^{+3} + SO_4^{-2}$

S-1

$6e^-$ (3)

$\overset{+6}{Cr_2O_7^{-2}} + \overset{+4}{SO_3^{-2}} \longrightarrow \overset{+3}{Cr} + \overset{+6}{SO_4^{-2}}$

$2e^-$ (1)

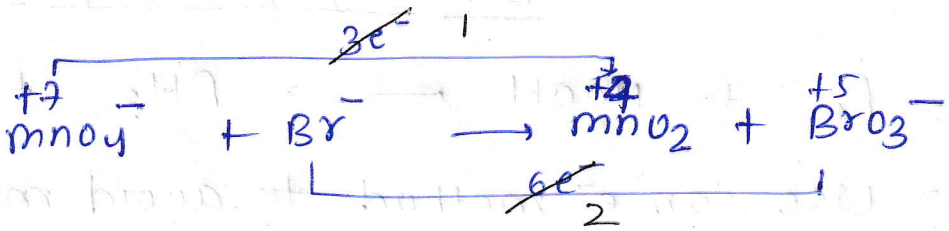
$Cr_2O_7^{-2} + 3SO_3^{-2} \longrightarrow Cr^{+3} + SO_4^{-2}$

S-2 $Cr_2O_7^{-2} + 3SO_3^{-2} \longrightarrow 2Cr^{+3} + 3SO_4^{-2}$

S-3 $8H^+ + Cr_2O_7^{-2} + 3SO_3^{-2} \longrightarrow 2Cr^{+3} + 3SO_4^{-2} + 4H_2O$

Basic ~~$MnO_4^- + Br^- \longrightarrow MnO_2 + BrO_3^-$~~

S1



S2 $2MnO_4^- + Br^- \longrightarrow 2MnO_2 + BrO_3^-$

S3 $2H^+ + 2MnO_4^- + Br^- \longrightarrow 2MnO_2 + BrO_3^- + H_2O$

S4 $2H^+ + 2MnO_4^- + Br^- \longrightarrow 2MnO_2 + BrO_3^- + H_2O + 2OH^-$

$2OH^-$

$2H_2O + 2MnO_4^- + Br^- \longrightarrow 2MnO_2 + BrO_3^- + H_2O + 2OH^-$

$2MnO_4^- + Br^- + H_2O \longrightarrow 2MnO_2 + BrO_3^- + 2OH^-$

~~Reaction~~

Balancing of disprop reaction

$$P_4 + NaOH \longrightarrow PH_3 + NaH_2PO_2$$

→ Use ion e^- method to avoid mistake.

S-1. Remove Spectator ion

$$P_4 + \text{[illegible]} \longrightarrow PH_3 + H_2PO_2^-$$

S-2 OH RH

$$\overset{0}{P_4} \longrightarrow \overset{-3}{PH_3} \qquad RH$$

$$\overset{0}{P_4} \longrightarrow \overset{+1}{H_2PO_2^-} \qquad OH$$

S-3 atom other than O & H

$$P_4 \longrightarrow 4PH_3$$

$$P_4 \longrightarrow 4H_2PO_2^-$$

S-4 O - H_2O, H - H^+

$$12H^+ + P_4 \longrightarrow 4PH_3$$

$$8H_2O + P_4 \longrightarrow 4H_2PO_2^- + 8H^+$$

Step 5 Balance Charge

$$12e^- + 12H^+ + \overset{0}{P_4} \longrightarrow \overset{-3}{4PH_3} \qquad \text{Redn}$$

$$8H_2O + \overset{0}{P_4} \longrightarrow \overset{+1}{4H_2PO_2^-} + 8H^+ + 4e^- \qquad \text{oxdn}$$

Step 6

$$12e^- + 12H^+ + P_4 \longrightarrow 4PH_3$$

$$24H_2O + 3P_4 \longrightarrow 12H_2PO_2^- + 24H^+ + 12e^-$$

$$24H_2O + 4P_4 + 12H^+ \longrightarrow 4PH_3 + 12H_2PO_2^- + 12H^+ \quad 12H_2O.$$

$12OH^-$ $\qquad$ $(12OH^-)$

Basic medium

$$12H_2O + 4P_4 + 12OH^- \longrightarrow 4PH_3 + 12H_2PO_2^- +$$

→ $12H_2O + 4P_4 + 12NaOH \rightarrow 4PH_3 + 12NaH_2PO_2$ → Take 4 common

n-factor

Valency factor / x factor

→ charge transfer per mole of a substance
or
no of e^- exchange per mole of a substance
} n factor.

Calculation of n-factor

① Ions

n-factor = |charge|

	Na^+	Mg^{+2}	Al^{+3}	SO_4^{-2}
x factor	1	2	3	2

② Salt

n factor = LCM of charge on cation & anion.

NaCl → (1)

MgO → (2)

$Al_2(SO_4)_3$ → (6)

③ acid / base

acid

n-factor = basicity

no of H^+ ion donated by acid

Basicity = 1	Basicity = 2	Basicity = 3
HCl	H_2SO_4	H_3PO_4
HBr	H_2CO_3	
HF	$H_2C_2O_4$	
$HClO_4$		
~~HNO3~~		
CH_3COOH		

base

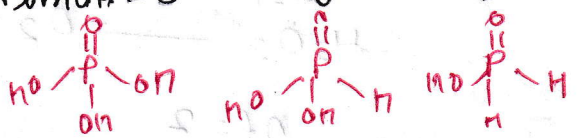
n-factor = acidity.

no of OH^- ion donate by base
or
no of H^+ ion accepted.

monoacidic	Diacidic	Triacidic
NaOH	$Mg(OH)_2$	$Al(OH)_3$
KOH	$Ca(OH)_2$	$Fe(OH)_3$
NH_4OH	$Zn(OH)_2$	
NH_3	$NH_2-CH_2-CH_2-NH_2$	
RNH_2		
$C_6H_5NH_2$		

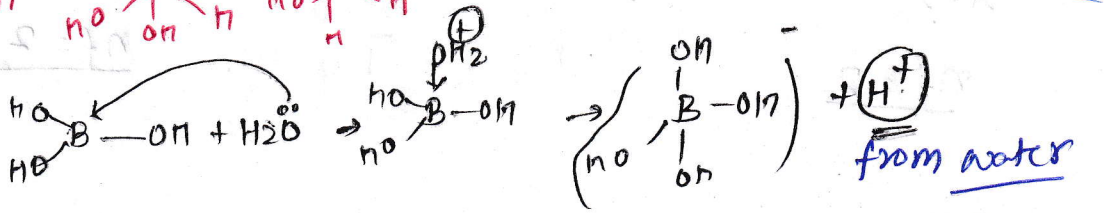
Phosphorous acids — H_3PO_4 H_3PO_3 H_3PO_2

Basicity = 3 2 1



Boric acid

H_3BO_3 / $B(OH)_3$



(4) For oxidising / Reducing agent

$KMnO_4$ → strong oxidising agent

MnO_4^- → acidic → Mn^{+2} nf = 5
basic → MnO_4^{-2} nf = 1
neutral → MnO_2 nf = 3

$Fe^{+2} \rightarrow Fe^{+3}$

$C_2O_4^{-2} \rightarrow CO_2$

$I^- \rightarrow I_2$

$H_2S \rightarrow S$

$H_2O_2 \rightarrow O_2$

$Sn^{+2} \rightarrow Sn^{+4}$

$SO_2 \rightarrow SO_4^{-2}$

$\underset{n\text{-}f=5}{MnO_4^-} + \underset{n\text{-}f=1}{Fe^{+2}} + H^+ \longrightarrow Mn^{+2} + Fe^{+3} + H_2O$

$\underset{n\text{-}f=5}{MnO_4^-} + \underset{nf=2}{C_2O_4^{-2}} + H^+ \longrightarrow Mn^{+2} + CO_2 + H_2O$

$\underset{5}{MnO_4^-} + \underset{1}{I^-} + H^+ \longrightarrow Mn^{+2} + I_2^0 + H_2O$

$\underset{5}{MnO_4^-} + \underset{2}{H_2S^{-2}} + H^+ \longrightarrow Mn^{+2} + S^0 + H_2O$

$\underset{5}{MnO_4^-} + \underset{2}{H_2O_2^{-1}} + H^+ \longrightarrow Mn^{+2} + O_2 + H_2O$

$\underset{5}{MnO_4^-} + \underset{2}{Sn^{+2}} + H^+ \longrightarrow Mn^{+2} + Sn^{+4} + H_2O$

$\underset{5}{MnO_4^-} + \underset{2}{SO_2} + H^+ \longrightarrow Mn^{+2} + SO_4^{-2} + H_2O$

Potassium dichromate

$Cr_2O_7^{-2} \xrightarrow{H^+} Cr^{+3}$ → nf = 6

$Fe^{+2} \rightarrow Fe^{+3}$

$C_2O_4^{-2} \rightarrow CO_2$

$I^- \rightarrow I_2$

$H_2S \rightarrow S$

$H_2O_2 \rightarrow O_2$

$Sn^{+2} \rightarrow Sn^{+4}$

$SO_2 \rightarrow SO_4^{-2}$

$\underset{6}{Cr_2O_7^{-2}} + \underset{1}{Fe^{+2}} + H^+ \longrightarrow Cr^{+3} + Fe^{+3} + H_2O$

$\underset{6}{Cr_2O_7^{-2}} + \underset{2}{C_2O_4^{-2}} + H^+ \longrightarrow Cr^{+3} + CO_2 + H_2O$

$\underset{6}{Cr_2O_7^{-2}} + \underset{1}{I^-} + H^+ \longrightarrow Cr^{+3} + I_2 + H_2O$

$Cr_2O_7^{-2} + \underset{2}{H_2S^{-2}} + H^+ \longrightarrow Cr^{+3} + S + H_2O$

$Cr_2O_7^{-2} + \underset{2}{H_2O_2^{-1}} + H^+ \longrightarrow Cr^{+3} + O_2^0 + H_2O$

$\underset{2}{Sn^{+2}} \longrightarrow Sn^{+4}$

$\underset{2}{SO_2} \longrightarrow SO_4^{+6\ -2}$

H_2O_2 → Agent as well as Ragent

H_2O_2 as oxidizing agent

$H_2O_2^{-1} \longrightarrow H_2O^{-2}$

nf = 2

H_2O_2 as reducing agent

$H_2O_2^{-1} \longrightarrow O_2^0$

nf = 2

Sodium thiosulphate $Na_2S_2O_3$ (Reducing agent)

$\overset{+2}{Na_2S_2O_3} \longrightarrow \overset{2.5}{Na_2S_4O_6}$

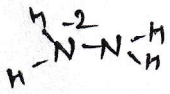
$nf = 0.5 \times 2 = 1$

tetrathionate

$nf = 0.5 \times 4 = 2$

one cal n-factor

$\overset{0}{I_2} \longrightarrow \overset{+5}{IO_3^-}$ $nf = 10$



$\overset{-2}{N_2H_4} \longrightarrow \overset{+3}{HNO_2}$ $nf = 10$

$\overset{+7}{IO_4^-} \longrightarrow \overset{0}{I_2}$ $nf = 7$

$\overset{+2}{VO} \longrightarrow \overset{+5}{V_2O_5}$ $nf = 3$

$Fe(\overset{+3}{C_2O_4})_3 \longrightarrow Fe^{+3} + \overset{+4}{CO_2}$ $nf = 6$

$\overset{+2}{Na_2S_2O_3} + \overset{0}{I_2} \longrightarrow \overset{+2.5}{Na_2S_4O_6} + 2\overset{-1}{NaI}$

$nf = 1$ $nf = 2$

When more than one oxidation state is changing

Case 1 : If oxidation num of elements either ↑ or ↓.

$\overset{+2}{Fe}\overset{+3}{C_2O_4} \longrightarrow Fe^{+3} + \overset{+4}{CO_2}$ $nf = \text{sum}$

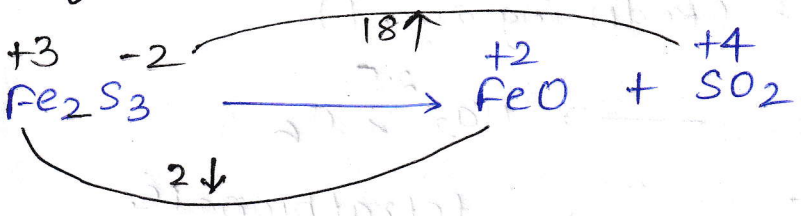
$\overset{+2}{FeS_2} \longrightarrow \overset{+3}{Fe_2O_3} + \overset{+4}{SO_2}$ $nf = 11$

$nf = 3$

$\overset{+3}{Fe}(\overset{+5}{NO_3})_3 \longrightarrow Fe^{+2} + \overset{+2}{NO}$ $nf = 10$

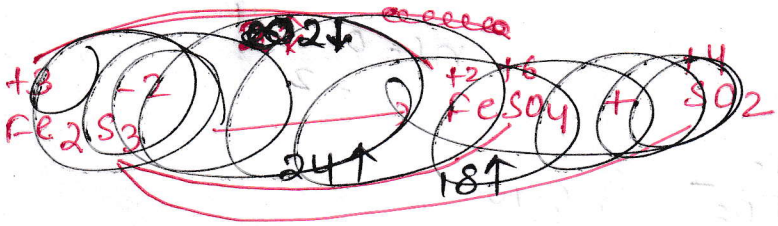
$\overset{+1}{Cu_2}\overset{-2}{S} \longrightarrow Cu^{+2} + \overset{+4}{SO_2}$ $nf = 8$

case2 → If ONum is ↑ and ↓.



$n_f = |n_1 - n_2|$

$= 16$

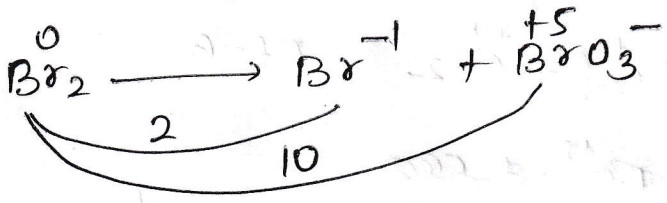


n-factor for disprop rxn

$$n_f = \frac{n_1 n_2}{n_1 + n_2}$$

→ Substance undergoing disprop.

Que



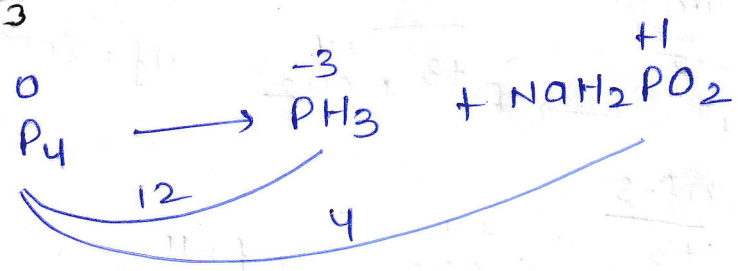
$n_f = 1$ $n_f = 5$

$n_f = \frac{2 \times 10}{2+10}$

$= \frac{20}{12}$

$= \frac{5}{3}$

Que



$n_f = 3$ $n_f = 1$

$n_f = \frac{12 \times 4}{12+4}$

$n_f = \frac{48}{16}$

$n_f = 3$

n-factor for balanced chemical reaction

$$n_f = \frac{\text{net } e^- \text{ exchange}}{\text{stoichiometry coeff}}$$

$$2\overset{+7}{K}\overset{+7}{Mn}O_4 + 5K_2\overset{+3}{C}_2O_4 + 16HCl \longrightarrow 2\overset{+2}{Mn}Cl_2 + 10\overset{+4}{C}O_2 + 12KCl + 8H_2O$$

(2 connects $\overset{+3}{C}$ to $\overset{+4}{C}$; 5 connects $\overset{+7}{Mn}$ to $\overset{+2}{Mn}$)

net e^- exchange = LCM (5, 2) = 10

$n_f = \frac{10}{2}$ | $n_f = \frac{10}{5}$ | $n_f = \frac{10}{16}$ | $n_f = \frac{10}{2}$ | $n_f = \frac{10}{10}$ | $n_f = \frac{10}{12}$ | $n_f = \frac{10}{8}$

Equivalent mass

$$\text{Eqt mass} = \frac{\text{molar mass}}{\text{n-factor}}$$

Que $\overset{0}{Mg} + \frac{1}{2}\overset{0}{O_2} \longrightarrow \overset{+2}{Mg}\overset{-2}{O}$

$n_f = 2$ | $n_f = 4$

$\text{Eq}^t\text{mass} = \frac{24}{2} = 12$ | $\text{Eq}^t\text{mass} = \frac{32}{4} = 8$

Que $\overset{0}{N_2} + 3\overset{0}{H_2} \longrightarrow 2\overset{-3}{N}\overset{+1}{H_3}$

$n_f = 6$ | $n_f = 2$

$\text{Eq mass} = \frac{28}{6}$ | $\text{Eq}^t\text{mass} = \frac{2}{2}$

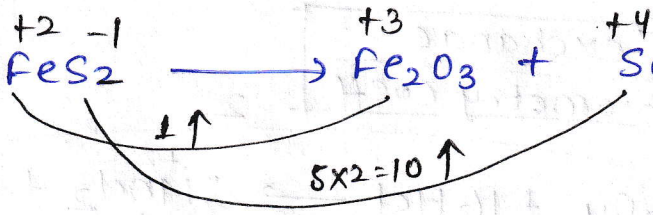
Que $\overset{+1}{H_3}PO_2 \longrightarrow \overset{-3}{P}H_3 + \overset{+3}{H_3}PO_3$

(4 connects $\overset{+1}{P}$ to $\overset{-3}{P}$; 2 connects $\overset{+1}{P}$ to $\overset{+3}{P}$)

$n_f = \frac{4\times2}{4+2} = \frac{8}{6} = \frac{4}{3}$

Eq mass of H_3PO_2 = $\frac{\text{Molar mass of } H_3PO_4}{4/3} = \frac{98\times3}{4\times2} = \frac{147}{2} = 73.5$ g

Que Cal Eq^t wt of FeS_2



$nf = 10+1 = 11$

$$\text{no of mole} = \frac{\text{Given mass}}{\text{molar mass}}$$

$$\text{no of Eq}^t = \frac{\text{Given mass}}{\text{Eq}^t \text{ mass}}$$

$$\text{no of Eq}^t = \frac{GM}{\frac{MM}{n\text{-}f}}$$

$$\text{no of Eq}^t = \frac{GM}{MM} \times n\text{-}f$$

$$\text{no of Eq}^t = \text{moles} \times n\text{-}f$$

Que Cal no of Eq^t in 31.6 gm $KMnO_4$ in acidic medium.

$$MnO_4^- \xrightarrow{H^+} Mn^{+2}$$

$nf = 5$

$$\text{no of Eq}^t = \text{mole} \times n\text{-factor}$$

$$= \frac{31.6}{158} \times 5$$

$$= 1$$

$$\text{Molarity} = \frac{\text{no of moles of solute}}{\text{Vol in litre solution}}$$

$$\text{no of moles of solute} = M \times \text{Vol (litre)}$$

$$\text{Normality} = \frac{\text{no of Eq}^t \text{ of solute}}{\text{vol of solution in litre}}$$

~~$N = \frac{n_{Eq}}{\text{Vol (litre)}}$~~

$$\text{No of Eq}^t \text{ of solute} = N \times \text{Vol (litre)}$$

Que Cal no of Eqt of NaOH in 50ml 0.5 N NaOH soln.

no of Eqt = N × Vol (litre)

$= \frac{0.5}{10} \times \frac{50}{1000}$

$= \frac{25}{1000}$

$= 0.025$ Eqt.

Eqt concept (no need to balance rxn)

Given N_2 = 56 gm Find wt of H_2 reacted

$N_2^0 + 3H_2^0 \longrightarrow 2\overset{-3}{N}\overset{+1}{H_3}$

no of Eqt of N_2 = $^{no\ of}$ Eqt of H_2 = $^{no\ of}$ Eqt of NH_3

moles of N_2 × n-f = moles of H_2 × n-f = moles of NH_3 × n-f

$\frac{56}{28} \times$ n-factor $= \frac{wt\ H_2}{2} \times$ n-factor

$2 \times 6 = \frac{(wt)\ H_2}{2} \times 2$

$(wt)\ H_2 = 12$ gm

Que Find wt of NH_3 formed by 56 gm N_2

$12 = \frac{(wt)\ NH_3}{17} \times$ n-factor

$12 \times 17 = Wt\ NH_3 \times (n_1 - n_2)$

$\times (3$

Sequential reaction

Find amount of H_2 reacted with 16gm O_2 to form H_2O

→ no need to balance rxn

$$\overset{0}{H_2} + \overset{0}{O_2} \longrightarrow \overset{+1\ \ -2}{H_2O}$$

?? 16gm

Eq⁺ of H_2 = Eq⁺ of O_2

$$\frac{Gm}{mm} \times n\text{-}f = \frac{Gm}{mm} \times n\text{-factor}$$

$$\frac{Gm}{2} \times 2 = \frac{16}{32} \times 4$$

$$Gm = 2gm$$